

Getting Data
PPDAC: Problem, Plan, Data, Analysis, and Conclusion

Variables
An attribute that can be measured or labelled.

- 1. Independent variable: may be subject to manipulation
- 2. Dependent variable: hypothesised to change

Variable classification
Needed to lead to correct visualisation tools (graphs)

- 1. Categorical Variable: like an enum type, each observation can be placed in 1 label
 - Ordinal: Natural ordering
 - Nominal: No natural ordering
- 2. Numerical Variable: like a number, manipulate with arithmetics
 - Discrete
 - Continuous

Sampling

Sample, Census, Population

- Sample: Sample is a proportion of population selected in the study.
- Census: Census is attempt to reach out to whole population.
- Population: Entire group of objects the researcher is interested in.

Sampling frame
A list of the items or people forming a population from which a sample is taken.

Goal: Sampling frame >= target population (generalisability -> sample is a good estimate of the population parameter; removes selection bias)

- Issues:
- 1. insufficient coverage
 - 2. redundant data

Bias

- 1. Selection Bias
 - Imperfect sampling frame: Parts of the population left out of the frame
 - Non-probability sampling: Omits chance in the selection of individuals (eg. arbitrarily selecting only women in a sample instead of randomly selecting)
- 2. Non-Response Bias

Non-disclosure of information toward the study due to reasons.

- Not interested
- Inconvenient
- Sensitive info

Probability, Non-probability sampling

- Probability: each unit has a known non zero chance of being selected (removes selection bias)
- Non-probability: not all units have a chance to be selected

Sampling methods

- 1. Simple random sampling
- 2. Stratified sampling: choose every *n*th unit
- 3. Stratified random sampling: divide into subgroups, perform random sampling on each subgroup. include selected units of subgroups in final sample
- 4. Cluster sampling: divide into subgroups, take random sample of subgroups. include all units of selected subgroups in final sample

Summary Statistics

- 1. Measures of central tendency
 - Mean
 - Median
 - Mode
- 2. Measures of Dispersion
 - Standard deviation
 - Inter-quartile range

Standard deviation and Variance
Sample variance: $Var(x) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
Sample s.d.: $s_x = \sqrt{Var(x)}$ (stdev.s in excel)

Coefficient of variation: $\frac{s_x}{\bar{x}}$
Median: middle value (if even numbered, take average of central values)
Inter-quartile range: $Q_1 - Q_3$

Marginal,

- marginal rate: (1 variable only, X) ratio of X and the overall population
- conditional rate: (2 variables, X | Y) the ratio of X and the overall population (taken as satisfying the condition Y)
- joint rate: (2 variables, X & Y) ratio where all the subjects considered the population (X & Y)

Study designs

Experimental studies
Manipulate independent variable to induce change in dependent variable. 1 treatment group, 1 control group. Shows cause and effect. Apply techniques:

- Random assignment: an impartial procedure that uses chance (to account for uncontrollable constants like age, sex, IQ, etc)
- Blinding: use a placebo to account for bias.
 - Blinded subjects do not know whether they are in the control or treatment group.
 - Blinded researchers do not know if the subject they are researching is in the control or treatment group.
- Double blind if both subjects and researchers are blinded

Observational studies
Researchers do not attempt to manipulate 1 variable to cause change in dependent variable. does not show cause or effect, instead shows association. used for ethical purposes.

example: is long term smoking linked to heart disease?

- Researchers should not force people to smoke
- They just observe the ‘treatment’ (those who smoke) and ‘control’ (those who don’t smoke) groups.

Categorical data analysis

Data representation

- Bar plots (can be percentage based)
 - Stacked (categorical groups are stacked atop each other to 100%)
 - Dodged (groups are placed beside each other)
- 2x2 table (binary outcomes + treatment + control, includes row and column totals)

Association between categorical variables
Assume A and B are characteristics of a population. *rate(X)* is the % of the population for which X is satisfied. If:

- *rate(A | B) = rate(A | ¬B)*: A is **not associated** with B. (rate is the same regardless of presence of B)
- *rate(A | B) > rate(A | ¬B)*: A has a **positive association** with B. (A and B together cause the rate to increase)
- *rate(A | B) < rate(A | ¬B)*: A has a **negative association** with B.

Example
Outcome of treatment:

- *A*: success
- *¬A*: failure

Treatment:

- *B*: Treatment X
- *¬B*: Treatment Y

rate(A | B) < rate(A | ¬B) means treatment X has a negative association with success and treatment Y has a positive association with success.

Rule on rates

Symmetry rule
 $rate(A | B) \circ rate(A | \neg B) \Leftrightarrow rate(B | A) \circ rate(B | \neg A)$ where $\circ \in \{=, <, >\}$

This gives results:

- 1. As $rate(B) \rightarrow 100\%$, $rate(A | B) \rightarrow rate(A)$
- 2. $rate(B) = 50\% \Rightarrow rate(A) = \frac{rate(A | B) + rate(A | \neg B)}{2}$
- 3. $rate(A | B) = rate(A | \neg B) \Rightarrow rate(A) = rate(A | B) = rate(A | \neg B)$

Basic rule
The *rate(A)* is always between subgroup rates *rate(A | B)* and *rate(A | C)*.

Simpson’s paradox
A trend appears in the majority of several groups of data, but disappears/reverses when the group is combined together.

- Every subgroup shows the same direction of association (e.g., A > B in male, A > B in female)
- But the combined data flips direction (overall B > A)

Use slicing to split by categorical groups, and use rules of rates to analyze. If sliced too many times, sample becomes inaccurate representation of population as it is too thin.

Eg. A doctor who treats more minor injuries than major injuries is likely to have a overall success rate than a doctor who does the other way around. But this may not be the case when splitting success rates between minor from major injuries.

Confounding variable
A third variable with an association with the dependant **and** independent variable.

Show confounding variable’s existence by proving $rate(C | X) \neq rate(C | \neg X)$.
Cannot possibly account for all confounders in an observational study. Hence association and not causation.
Correct for confounders in experimental studies with random assignment.

Presence of Confounders != presence of Simpson’s paradox BUT presence of Simpson’s paradox === presence of confounders

Dealing with numerical data

Univariate exploratory data analysis (EDA)
Describe: shape, centre, spread, outliers

- frequency table
- histogram: divide data into ‘bins’ (segments, eg 1-2, 3-4, 5-6)
 - Unimodal/Multimodal: 1 peak/many peaks in the data
 - Symmetrical/(L/R)-skewed: Peak is in the centre/skewed to the left or right
 - Mean, median, mode: Positions depend on skew, R: mean < median < mode; L: mode < median < mean
 - Variability (s.d., range of data)
- box plot: draw a box from Q1 to Q3 + median, with whiskers from Q1 - 1.5 * IQR to Q3 + 1.5 * IQR
 - 5-number summary: Q0-Q4
 - outliers (indicate with dots): data points which are
 - 1. greater than Q3 + 1.5 * IQR
 - 2. less than Q1 - 1.5 * IQR

Bivariate EDA
Deterministic relationship: where the dependent variable can be directly calculated from the independent variable

- Scatter plots
 - ordered pair (x,y) to scatter points on a graph
 - relationship: positive, negative, NA
 - form: linear, non-linear
 - strength of relationship: strong (points follow relationship closely) vs weak (vice-versa)
- Correlation coefficients
- Regression analysis

Linear regression

- Corelation coefficient *r*, $-1 \leq r \leq 1$
 - sign of r: positive/negative linear association
 - magnitude of r: strength of linear association
 - passes through $(\bar{x}, \bar{y})$
 - standard unit (SU) $= \frac{x-\bar{x}}{s_x}$
 - $r = \frac{1}{n-1} \sum \left(\frac{x-\bar{x}}{s_x} \right) \left(\frac{y-\bar{y}}{s_y} \right)$
 - $m = r \left(\frac{s_y}{s_x} \right)$
 - not affected by:
 - 1. switching of x and y
 - 2. adding constant to x or y
 - 3. multiplying by positive constant
- normalisation: $Y = mX + b$, take r value with highest magnitude.
- ecological correlation: aggregate data points into groups, analyse regression separately

Probability

- **Probability experiment**: MUST be
 - 1. repeatable
 - 2. give rise to a precise set of outcomes
- **Sample space**: collection of all outcomes
- **Event**: subcollection of sample space (outcome is an event)
- **uniform probability**: every outcome has same probability

$P(A|B)$ = rate(*A|B*) (but rates do not involve random processes, rate is the actual result)
 $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Bayes theorem: $P(A | B) = \frac{P(B | A)P(A)}{P(B)}$
Independent events: $P(A) \times P(B) = P(A \cap B)$ (the occurrence of 1 event does not change the likelihood of the other event)

Mutually exclusive events: $P(A \cup B) = P(A) + P(B)$

Law of total probability: total probability of an event (A) can be found by summing the probabilities of (A) occurring within each of a set of mutually exclusive and exhaustive events $(B_1, B_2, ..., B_n)$, ie. $P(A) = \sum P(A|B_i)P(B_i)$

Fallacies

Prosecutor’s fallacy: $P(A|B) = P(B|A)$, which is false.

Conjunction fallacy: $P(A \cap B) > P(A)$, which is false.

Base rate fallacy: A cognitive bias where people ignore statistical “base rates” (the general prevalence of an event or trait) and instead focus on specific, individuating information, leading to inaccurate probability judgments.

ecological fallacy: the mistake of assuming that characteristics or relationships observed at a group level (like average scores for a city) also apply to individuals within that group. may yield wildly different r values.

atomistic fallacy: inferences about a higher-level unit (such as a group) are made based on data collected at a lower level (such as individuals). Opposite of ecological fallacy

Statistical inference

Use a sample to draw a conclusion of the population.

sample statistic = population parameter + random error + bias

Confidence interval

an interval of values computed from sample data that is likely to include the unknown value of a population parameter. expressed as sample statistic $\pm$ margin of error

formula: sample statistic $\pm$ multiplier $\times$ standard error

- multiplier: a number based on on the confidence level desired
- standard error: estimate of st. deviation

eg: 95% CI: 1.2 ± 0.1 or $[1.3, 1.5]$ => if many simple random samples are obtained, and a confidence level is constructed for each of them, 95% of the confidence intervals constructed would be in the population parameter (the actual value of the question posed)

Chi Squared test

1. establish null and alternative hypothesis
2. collect sample data
3. calculate expected data if null hypothesis is true
4. get chi-squared value: $\chi^2 = \sum \frac{(O_i - E_i)^2}{O_i}$
5. if chi-squared value is large, p-value will be small

NOTE: only use when generalising a sample to be applicable to the population. otherwise, use $\text{rate}(A|B) == \text{rate}(A| \neg B)$

Z Test (Population Mean, σ Known)

1. State the hypotheses: $H_0: \mu = \mu_0$, with one- or two-sided H_a .
2. Collect sample data: sample mean $\bar{x}$, known σ , sample size n.
3. Compute the test statistic:

$$z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}}$$

4. Compute the p-value using the standard normal distribution.
5. Compare the p-value to α and decide whether to reject H_0 .

T Test (Population Mean, σ Unknown)

1. State the hypotheses: $H_0: \mu = \mu_0$, with one- or two-sided H_a .
2. Collect sample data: sample mean $\bar{x}$, sample standard deviation s, sample size n.
3. Compute the test statistic:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

Degrees of freedom: $df = n - 1$.

4. Compute the p-value using the t-distribution with df degrees of freedom.
5. Compare the p-value to α and decide whether to reject H_0 .

Excel Functions Reference

Statistical Functions

AVERAGE()

Arguments: number1, number2, ... (numbers or ranges)

Returns: Number

Description: Calculates the arithmetic mean of a set of values. Ignores text, logical values, and empty cells.

Example: =AVERAGE(A1:A10) returns the mean of values in cells A1 through A10.

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MEDIAN()

Arguments: number1, number2, ... (numbers or ranges)

Returns: Number

Description: Returns the middle value in a dataset when sorted. If the dataset has an even number of values, returns the average of the two middle values.

Example: =MEDIAN(B1:B20) returns the median value from the range B1 to B20.

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MODE()

Arguments: number1, number2, ... (numbers or ranges)

Returns: Number

Description: Returns the most frequently occurring value in a dataset. Returns #N/A error if no value appears more than once.

Example: =MODE(C1:C15) finds the most common value in the range.

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STDEV.S()

Arguments: number1, number2, ... (numbers or ranges)

Returns: Number

Description: Calculates the sample standard deviation, which estimates the standard deviation based on a sample of the population. Uses the “n-1” method.

Example: =STDEV.S(D1:D50) calculates sample standard deviation for the dataset.

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MAX()

Arguments: number1, number2, ... (numbers or ranges)

Returns: Number

Description: Returns the largest value in a set of values. Ignores text and logical values.

Example: =MAX(E1:E100) finds the maximum value in the range.

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MIN()

Arguments: number1, number2, ... (numbers or ranges)

Returns: Number

Description: Returns the smallest value in a set of values. Ignores text and logical values.

Example: =MIN(F1:F100) finds the minimum value in the range.

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CORREL()

Arguments: array1 (range), array2 (range)

Returns: Number (between –1 and 1)

Description: Calculates the Pearson correlation coefficient between two datasets, measuring the strength and direction of the linear relationship between them.

Example:

=CORREL(A1:A20, B1:B20)

Returns the correlation between two variables. A value of 1 indicates perfect positive correlation, –1 indicates perfect negative correlation, and 0 indicates no linear correlation.

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SLOPE()

Arguments: known_y's (range), known_x's (range)

Returns: Number

Description: Calculates the slope of the linear regression line through the given data points using least squares method. Returns the rate of change of y with respect to x.

Example:

=SLOPE(B2:B15, A2:A15)

If A2:A15 contains years and B2:B15 contains sales figures, this returns the annual rate of change in sales. For instance, a result of 5000 means sales increase by 5,000 per year on average.

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CONFIDENCE.T()

Arguments: alpha (number), standard_dev (number), size (number)

Returns: Number

Description: Calculates the confidence interval using Student’s t-distribution. Used when sample size is small or population standard deviation is unknown. Returns the margin of error value.

Example:

=CONFIDENCE.T(0.05, 2.5, 30)

Calculates the 95% confidence interval (alpha=0.05) for a sample with standard deviation 2.5 and sample size 30. If the sample mean is 50, the confidence interval would be 50 $\pm$ result.

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COUNT()

Arguments: value1, value2, ... (values or ranges)

Returns: Number (integer)

Description: Counts the number of cells containing numeric values. Ignores empty cells, text, logical values, and errors.

Example: =COUNT(A1:A100) returns how many cells in the range contain numbers.